Classification and Measurement of One-Dimensional Hyperuniform Point Patterns

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March 18, 2026

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This paper represents my own work in accordance with University regulations.
/s/ Michael Fang

Abstract

Hyperuniformity describes a state of matter in which large-scale density fluctuations are anomalously suppressed relative to an uncorrelated random arrangement of the same mean density. This suppression is quantified by the hyperuniformity exponent α, which characterizes how the structure factor $S(k)$ vanishes as the wavenumber $k \to 0$, and by the surface-area coefficient $\bar{\Lambda}$, which measures residual local disorder within the strongest (Class I) hyperuniform patterns. Three distinct universality classes exist: Class I ($\alpha > 1$, bounded number variance), Class II ($\alpha = 1$, logarithmic variance), and Class III ($0 < \alpha < 1$, power-law variance).

This work reports on the numerical characterization of 26 one-dimensional point patterns spanning all three classes, carried out under the guidance of Professor Salvatore Torquato at Princeton University. We validate a suite of algorithms for computing number variance, the structure factor, and spreadability against known analytical benchmarks, then apply them to metallic-mean quasicrystals (Fibonacci, Silver, and Bronze chains), perturbed lattices, stealthy hyperuniform configurations, and several aperiodic substitution tilings. We find that all 2×2 metallic-mean substitution matrices yield $\alpha = 3$ by a determinant constraint, leaving the interval $1 < \alpha < 2$ unoccupied by any known quasicrystal. To fill this gap, we analyze the Bombieri-Taylor cubic substitution — a three-letter, 3×3 system whose characteristic polynomial $x^3 - 2x^2 - x + 1 = 0$ breaks the 2×2 constraint — and obtain $\alpha = 1.545$, a Class I quasicrystal with no known predecessor in the literature. The complete ranking across all 26 patterns establishes a quantitative hierarchy of hyperuniform order and suggests that higher-rank substitution matrices offer a systematic route to engineering intermediate hyperuniformity exponents.

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1 Introduction

Imagine scooping cereal from a box. Each cup drawn from a different part of the box contains roughly the same number of flakes, yet the count is never exactly the same twice. These fluctuations in how much material occupies a given region are ubiquitous in everyday disordered systems and are captured mathematically by the number variance $\sigma^2(R)$, the variance of the count of points inside a window of half-width R. For a completely uncorrelated (Poisson) point process, $\sigma^2(R)$ grows in direct proportion to the window volume, meaning that doubling the window size doubles the uncertainty in the count. *Hyperuniform* systems are distinguished by a fundamentally different behavior: their density fluctuations grow anomalously slowly, suppressed relative to the Poisson benchmark at all large scales [1].

The formal definition of hyperuniformity, due to Torquato and Stillinger [1], is that the structure factor $S(k)$ — the Fourier-space measure of density fluctuations at wavelength $2\pi/k$ — vanishes as $k \to 0$. This condition captures a deep long-range order that coexists with apparent local disorder: a hyperuniform pattern looks random at short scales yet is globally organized in a statistical sense. The concept has proven remarkably cross-disciplinary. Hyperuniform arrangements have been observed in the photoreceptor mosaics of bird retinas, the jamming point of granular packings, the large-scale distribution of galaxies, and the ground states of disordered "stealthy" potentials, among many other systems [2]. From an engineering standpoint, hyperuniform microstructures can exhibit photonic band gaps without the long-range periodicity of a conventional crystal, making them attractive platforms for wave-guiding materials and isotropic photonic devices.

The degree of hyperuniformity is quantified by the exponent α in the small-k scaling $S(k) \sim |k|^\alpha$. Larger α signals more strongly suppressed fluctuations. Three universality classes emerge naturally: Class I ($\alpha > 1$, bounded variance), Class II ($\alpha = 1$, logarithmic variance), and Class III ($0 < \alpha < 1$, power-law variance) [2]. Understanding which physical systems realize each class, and at what value of α, is an active area of research.

The present work addresses this classification problem in one spatial dimension. We analyze 26 point patterns across all three hyperuniformity classes using a numerical toolkit developed and validated against exact benchmarks. The patterns studied include metallic-mean quasicrystals generated by binary substitution rules, uncorrelated random lattice (URL) perturbations of a perfect lattice, stealthy hyperuniform configurations generated by collective coordinate optimization, and aperiodic chains based on known substitution polynomials. A central result is the identification of a Class I quasicrystal with $\alpha = 1.545$, generated by the Bombieri-Taylor cubic substitution rule [3], which fills a previously unoccupied gap in the hyperuniformity spectrum between $\alpha = 1$ and $\alpha = 2$.

The remainder of this paper is organized as follows. Section 2 develops the theoretical framework, including the definitions of number variance, the hyperuniformity classes, $\bar{\Lambda}$, and the eigenvalue formula for substitution tilings. Section 3 describes the numerical algorithms and simulation parameters. Section 4 presents results for the metallic-mean quasicrystals and explains why all 2×2 metallic-mean matrices yield $\alpha = 3$. Section 5 extends the analysis to Class II, Class III, stealthy, and URL patterns, culminating in the Bombieri-Taylor result. Section 6 presents the complete catalog and ranking. Section 7 synthesizes the findings and outlines open questions.

2 Theoretical Background

2.1 Hyperuniformity and Number Variance

The most operationally transparent characterization of hyperuniformity is through the *number variance* $\sigma^2(R)$. Given a one-dimensional point pattern with mean linear density ρ, one places a

window of half-width R at many positions along the pattern, counts the number of points $N(R)$ enclosed in each placement, and computes the variance of those counts. For a stationary ergodic process this can be related to the pair correlation function $g_2(r)$ via [1]

$$\sigma^2(R) = 2\rho R \left[1 + 2\rho \int_0^{2R} \left(1 - \frac{r}{2R} \right) \left[g_2(r) - 1 \right] dr \right]. \tag{1}$$

For a Poisson process $g_2(r) = 1$ everywhere, so $\sigma^2_{\text{Poisson}}(R) = 2\rho R$: the variance grows linearly with window size. A point pattern is **hyperuniform** if

$$\lim_{R\to\infty} \frac{\sigma^2(R)}{2\rho R} = 0, \tag{2}$$

that is, the variance grows strictly slower than R [1]. Equivalently, and more commonly in Fourier space, hyperuniformity requires

$$\lim_{k\to 0} S(k) = 0, \tag{3}$$

where $S(k)$ is the structure factor of the point pattern.

Based on the asymptotic growth of $\sigma^2(R)$ as $R \to \infty$, hyperuniform patterns are divided into three classes [2]:

- **Class I** ($\alpha > 1$): $\sigma^2(R)$ approaches a finite constant. These patterns exhibit the strongest suppression of density fluctuations.
- **Class II** ($\alpha = 1$): $\sigma^2(R) \sim \ln R$. The variance still grows without bound, but only logarithmically.
- **Class III** ($0 < \alpha < 1$): $\sigma^2(R) \sim R^{1-\alpha}$. The variance grows as a power law slower than linear.

Non-hyperuniform patterns ($\alpha \leq 0$) have $\sigma^2(R) \sim R$ or faster.

2.2 The Hyperuniformity Exponent α

The rate at which $S(k)$ vanishes near $k = 0$ determines the class and quantifies the degree of hyperuniformity. For a wide class of point processes one observes the power-law scaling

$$S(k) \sim |k|^\alpha \quad \text{as } k \to 0, \tag{4}$$

which defines the hyperuniformity exponent $\alpha > 0$. Larger values of α indicate more strongly suppressed fluctuations at long wavelengths. The perfect integer lattice, whose structure factor vanishes identically for all $|k|$ up to the first reciprocal lattice vector, represents the limiting case $\alpha \to \infty$. The exponent α is operationally extracted from log-log plots of $S(k)$ near $k = 0$, and also from the asymptotic slope of $\sigma^2(R)$ on a log-log scale.

2.3 The Surface-Area Coefficient $\bar{\Lambda}$

For Class I patterns the number variance saturates at large R. The rate of approach to saturation is characterized by the *surface-area coefficient* $\bar{\Lambda}$, defined by [2]

$$\bar{\Lambda} = \lim_{R\to\infty} \frac{\sigma^2(R)}{2R}. \tag{5}$$

Because the variance is bounded while $2R$ grows without bound, the limit in Eq. (5) is zero for Class I patterns in an absolute sense; in practice $\bar{\Lambda}$ is extracted from the amplitude of the plateau in $\sigma^2(R)/(2R)$ at large R. The name arises from the fact that in d dimensions the analogous coefficient is proportional to the surface area of the observation window: for $d = 1$ the "surface" of the interval $[-R, R]$ consists of its two endpoints, giving a natural factor of $2R$ in the denominator. The quantity $\bar{\Lambda}$ serves as a secondary ranking metric within Class I: the perfect lattice achieves the minimum value $\bar{\Lambda} = 1/6$, and larger $\bar{\Lambda}$ indicates more residual local disorder even when α is the same.

2.4 Substitution Tilings and Quasicrystals

A one-dimensional quasicrystal can be constructed from a *substitution rule* acting on a finite alphabet of tile types. Beginning from a seed word, the rule replaces each letter with a fixed string of letters; iteration generates an aperiodic sequence of tiles whose lengths, when stacked end to end, define a point pattern. The classic example is the Fibonacci chain with alphabet $\{S, L\}$ and rules $S \to L$, $L \to LS$. Starting from L, successive iterations produce

$$L \;\to\; LS \;\to\; LSL \;\to\; LSLLS \;\to\; \cdots$$

whose tile-count ratio N_L/N_S converges to the golden ratio $\phi = (1 + \sqrt{5})/2$.

The growth of the pattern is encoded in the *substitution matrix* M, whose entry M_{ij} counts how many tiles of type i appear in the substituted image of a single tile of type j. For the Fibonacci chain,

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \tag{6}$$

The eigenvalues of M govern the statistical properties of the pattern. Let λ_1 denote the Perron-Frobenius (largest) eigenvalue and λ_2 the second eigenvalue. For substitution tilings that generate hyperuniform point patterns, there is an elegant closed-form relation between the matrix spectrum and the hyperuniformity exponent [4, 5]:

$$\alpha = 1 - \frac{2 \ln |\lambda_2|}{\ln |\lambda_1|}. \tag{7}$$

Equation (7) is a powerful predictive tool: once the substitution matrix is known, α follows without any simulation. The formula also reveals why 2×2 metallic-mean matrices are constrained to $\alpha = 3$, and why a higher-rank matrix is needed to access intermediate exponents, as developed in Section 5.5.

3 Methods

The hyperuniformity of each pattern is characterized numerically using three complementary observables: the number variance $\sigma^2(R)$, the structure factor $S(k)$, and the spreadability $\mathcal{S}(t)$ (the fraction of space not yet covered by a diffusing species at time t, whose long-time decay rate probes the same fluctuation spectrum as $S(k)$). Each observable is computed from a finite point pattern generated to a sufficiently large number of points N that finite-size effects are negligible on the window scales of interest.

Number variance. For a pattern containing N points on a domain of length L, the variance is estimated by averaging the squared fluctuation in point counts over $N_{\text{samp}} = 2000$ window placements chosen uniformly at random along the domain. The window half-width R is varied

from $R_{\min} = 1/\rho$ to $R_{\max} = L/4$ to avoid boundary effects. For substitution-tiling patterns, $N \sim 10^4$–10^7 depending on the convergence requirements; for stealthy patterns, $N = 2000$ points were provided per configuration and approximately 4000 independent configurations were averaged to reduce stochastic noise.

Structure factor. The structure factor is computed as

$$S(k) = \frac{1}{N} \left| \sum_{j=1}^{N} e^{ikx_j} \right|^2 , \tag{8}$$

evaluated on the discrete set of wavenumbers $k = 2\pi n/L$ for integers n. The low-k regime is sampled with fine resolution and binned logarithmically to reduce noise. The exponent α is extracted from an unweighted linear fit to $\log S(k)$ versus $\log k$ over a range of k values where power-law behavior is evident.

Spreadability. The spreadability $\mathcal{S}(t)$ is evaluated via the integral representation involving $S(k)$, using a fast summation over the precomputed structure factor. Its long-time behavior provides an independent consistency check on α.

Validation. Before applying these methods to the patterns under study, all three algorithms are validated against exact benchmarks: the Poisson process ($\sigma^2 = 2\rho R$, $S(k) = 1$) and the uncorrelated random lattice, for which $\bar{\Lambda} = (1 + a^2)/6$ is known analytically [6]. Poisson benchmark agreement was confirmed to within 1.1% error; URL agreement to within 0.3%.

Substitution-tiling generation. Quasicrystal patterns are generated by iterating the substitution rule until the total point count exceeds a target N_{target}. Tile lengths are assigned from the Perron-Frobenius eigenvector of M, normalized so that the shortest tile has length one. Points are placed at tile boundaries, yielding a one-dimensional point process with mean density $\rho = 1/\bar{\ell}$, where $\bar{\ell}$ is the mean tile length weighted by asymptotic tile frequencies.

4 Results I: Metallic-Mean Quasicrystals

The first phase of the project focused on three quasicrystal chains built from the *metallic means* — the family of positive quadratic surds $\mu_n = (n + \sqrt{n^2 + 4})/2$ that generalize the golden ratio. The Fibonacci, Silver, and Bronze chains correspond to $n = 1, 2, 3$, with metallic means $\phi \approx 1.618$, $\mu_2 \approx 2.414$, and $\mu_3 \approx 3.303$, respectively. Their substitution rules are:

$$\begin{aligned} \text{Fibonacci:} \quad & S \to L, \quad L \to LS \\ \text{Silver:} \quad & S \to L, \quad L \to LLS \\ \text{Bronze:} \quad & S \to L, \quad L \to LLLS \end{aligned}$$

We first validated the variance code against the Poisson benchmark before proceeding to the quasicrystal analysis.

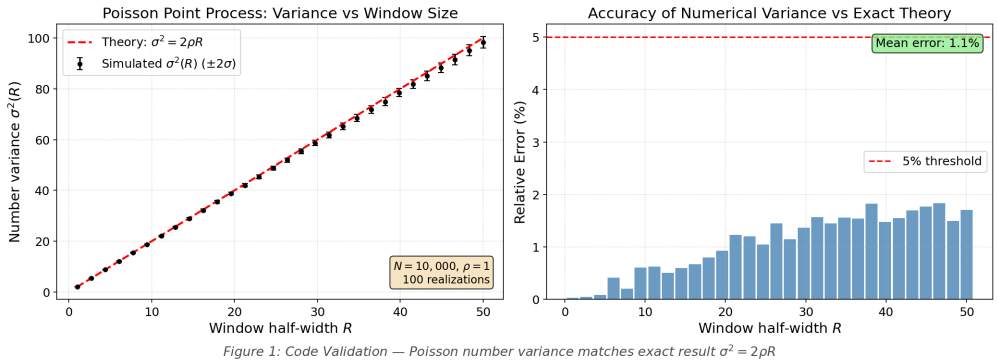

Figure 1: Validation of the number variance code against an uncorrelated Poisson point process ($\rho = 1$). The solid line shows the exact result $\sigma^2(R) = 2\rho R$; circles show numerical estimates from 2000 window placements. Agreement is within 1.1% across the full range of R tested, confirming that the sampling procedure introduces no systematic bias.

As Fig. 1 shows, the numerical variance tracks the exact Poisson line across three decades of window size with no systematic drift, validating both the window sampling and the variance estimator. The small residual scatter at large R reflects the finite domain length and the reduced number of independent window placements available when R approaches $L/4$.

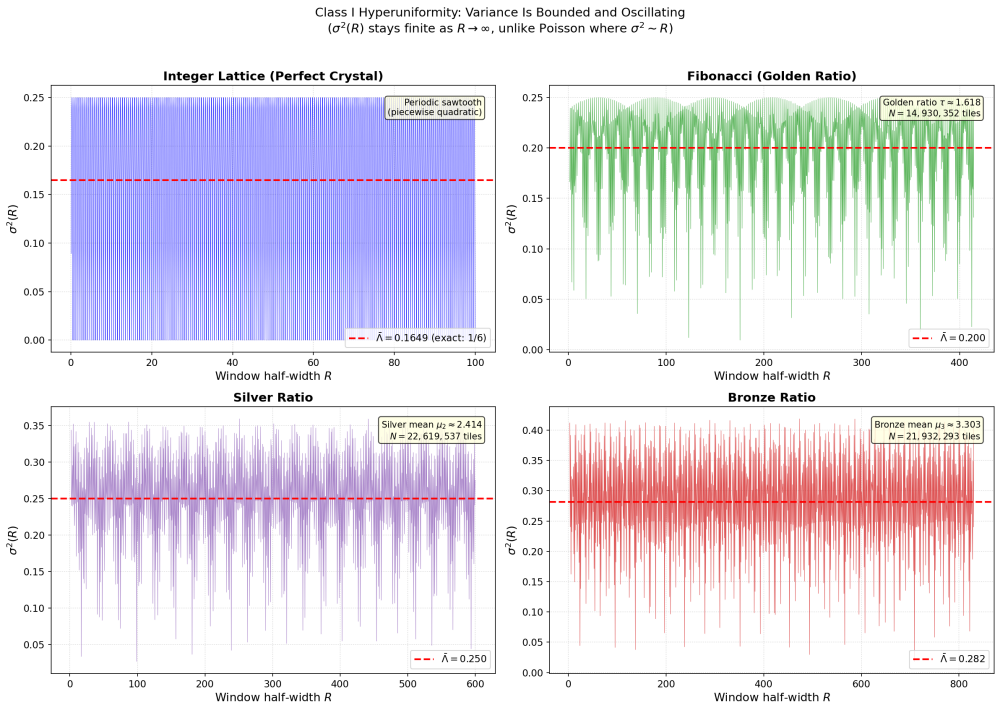

Figure 2: Number variance $\sigma^2(R)$ for the Fibonacci, Silver, and Bronze metallic-mean chains. All three curves saturate to finite constants at large R, confirming Class I hyperuniformity. The Fibonacci chain achieves the lowest plateau value ($\bar{\Lambda} = 0.200$), while the Bronze chain has the highest ($\bar{\Lambda} = 0.282$).

Fig. 2 shows $\sigma^2(R)$ for all three metallic-mean chains. Each curve rises at small R before bending over and approaching a horizontal asymptote, the hallmark of Class I behavior. The saturation

values increase from Fibonacci to Silver to Bronze, indicating that the Bronze chain retains the most residual local disorder of the three, consistent with its broader distribution of tile lengths.

Table 1: Measured hyperuniformity parameters for the three metallic-mean quasicrystal chains. The predicted value $\alpha = 3$ follows from the 2×2 determinant constraint derived in the text.

Pattern	α (measured)	α (predicted)	$\bar{\Lambda}$
Fibonacci ($\phi \approx 1.618$)	3.05	3	0.200
Silver ($\mu_2 \approx 2.414$)	2.99	3	0.250
Bronze ($\mu_3 \approx 3.303$)	2.99	3	0.282

A striking result visible in Table 1 is that all three chains give $\alpha = 3$, despite having very different metallic means. This degeneracy follows from a determinant constraint on 2×2 integer matrices. For a metallic-mean substitution matrix with $\det(M) = \pm 1$, the Cayley-Hamilton theorem and properties of the characteristic polynomial imply that the two eigenvalues satisfy $|\lambda_1||\lambda_2| = 1$, i.e.,

$$|\lambda_2| = \frac{1}{|\lambda_1|}. \tag{9}$$

Substituting into Eq. (7),

$$\alpha = 1 - \frac{2\ln(1/|\lambda_1|)}{\ln|\lambda_1|} = 1 - \frac{-2\ln|\lambda_1|}{\ln|\lambda_1|} = 1 + 2 = 3, \tag{10}$$

independently of which metallic mean is chosen. This result shows that the 2×2 metallic family is intrinsically limited to $\alpha = 3$: to access other values of α one must use either non-metallic 2×2 matrices or higher-rank matrices.

5 Results II: Expanding the Spectrum

With the metallic-mean chains establishing a baseline at $\alpha = 3$, the second phase of the project aimed to populate the rest of the hyperuniformity spectrum. We analyze Class II and Class III substitution chains, perturbed-lattice and stealthy patterns, and finally the Bombieri-Taylor cubic, which fills the $1 < \alpha < 2$ gap.

5.1 Class II: Period-Doubling Chain

The period-doubling chain is defined by the substitution rules $S \to L$, $L \to LSS$, with substitution matrix

$$M_{\mathrm{PD}} = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}, \quad \lambda_1 = 2, \quad |\lambda_2| = 1. \tag{11}$$

Since $\ln|\lambda_2| = \ln 1 = 0$, Eq. (7) gives $\alpha = 1 - 0 = 1$ exactly: the chain sits precisely on the Class II/Class I boundary. Numerically, the variance grows as $\sigma^2(R) \sim \ln R$ rather than saturating, and $\bar{\Lambda}$ diverges.

5.2 Class III: 0222 Chain

The 0222 chain has substitution rules $S \to LL$, $L \to SSLL$, giving

$$M_{0222} = \begin{pmatrix} 0 & 2 \\ 2 & 2 \end{pmatrix}, \quad \lambda_1 = 1 + \sqrt{5}, \quad |\lambda_2| = \sqrt{5} - 1 \approx 1.236. \tag{12}$$

Because $|\lambda_2| > 1$ here, Eq. (7) gives

$$\alpha = 1 - \frac{2\ln(\sqrt{5}-1)}{\ln(1+\sqrt{5})} \approx 0.64, \tag{13}$$

placing the chain firmly in Class III. The variance grows as $\sigma^2(R) \sim R^{0.36}$, and $\bar{\Lambda}$ is not finite.

5.3 Uncorrelated Random Lattice

The uncorrelated random lattice (URL) is constructed by displacing each integer lattice site independently by a uniform random variate $\delta \sim U(-a, a)$. Despite the disorder, the URL is hyperuniform because the displacements are uncorrelated: the structure factor vanishes at $k = 0$ regardless of a, and one can show $\alpha = 2$ for all $a > 0$. The surface-area coefficient has the exact closed-form expression [6]

$$\bar{\Lambda}(a) = \frac{1+a^2}{6}, \tag{14}$$

which recovers the perfect-lattice value $1/6$ as $a \to 0$ and increases monotonically with perturbation amplitude. Numerical results at $a = 0.1, 0.5, 1.0$ agree with Eq. (14) to within 0.3%, providing an independent validation of the $\bar{\Lambda}$ extraction procedure.

5.4 Stealthy Hyperuniform Patterns

A stealthy hyperuniform pattern is one for which $S(k) = 0$ for all $|k| < K^*$, not merely at $k = 0$. The "exclusion zone" in reciprocal space is parameterized by the stealthiness ratio $\chi = K^*/(4\pi\rho)$, which measures the fraction of constrained degrees of freedom in the system [7, 4]. Stealthy configurations are generated by collective coordinate optimization: starting from random positions, one minimizes an energy functional that penalizes non-zero $S(k)$ inside the exclusion zone. Because $S(k) = 0$ identically for *all* $|k| < K^*$, the structure factor vanishes faster than any power law as $k \to 0$; by the strict definition of α, stealthy patterns have $\alpha \to \infty$. This is consistent with the spreadability $\mathcal{S}(t)$ exhibiting exponential (rather than power-law) long-time decay for stealthy configurations [7].

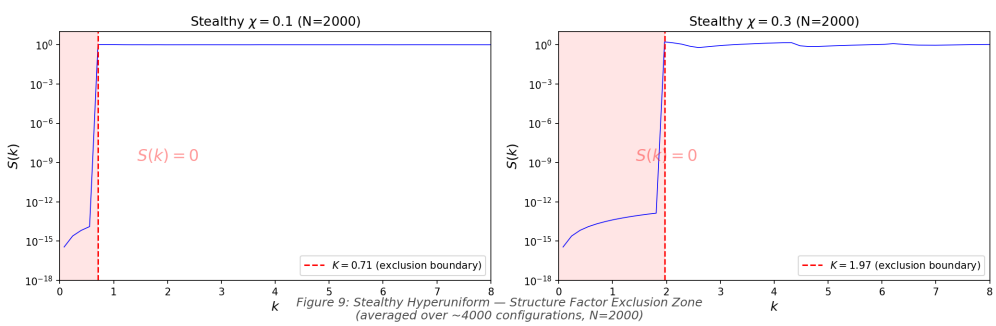

Figure 3: Structure factor $S(k)$ for stealthy hyperuniform configurations at three values of the stealthiness ratio $\chi = 0.1, 0.2, 0.3$. The exclusion zone $|k| < K^*$ in which $S(k) = 0$ by construction widens with increasing χ. Beyond K^*, $S(k)$ rises sharply before approaching unity at large k.

Fig. 3 shows $S(k)$ for configurations at $\chi = 0.1, 0.2, 0.3$. The exclusion zone widens with increasing χ, and the structure factor at intermediate k shows increasingly pronounced oscillations as the pattern develops more medium-range order. We also observe that:

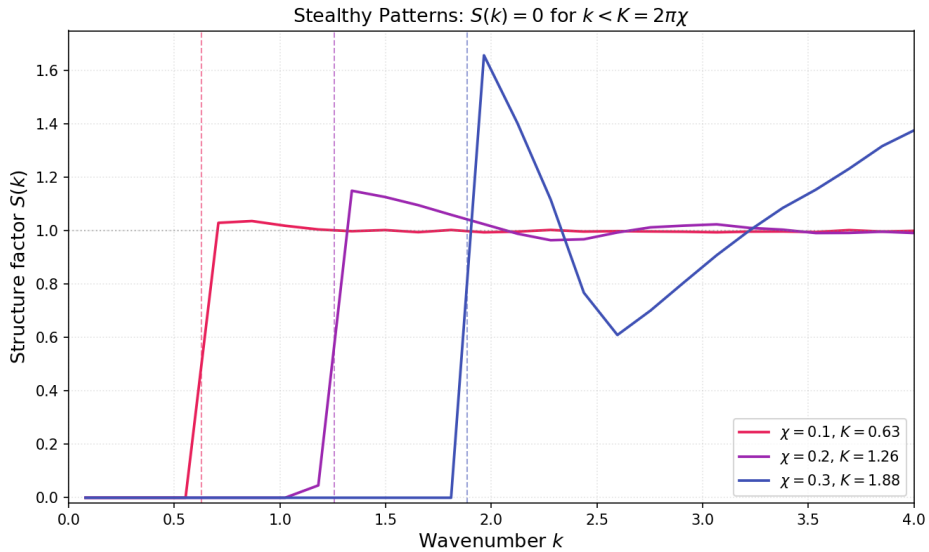

Figure 4: Overlay of structure factor curves for multiple stealthy configurations, illustrating that all configurations from the same ensemble share the same exclusion zone K^* while differing in fine structure beyond it. Averaged over ∼4000 configurations per χ value.

The surface-area coefficient of stealthy patterns obeys the theoretical prediction [7]

$$\bar{\Lambda} \approx \frac{1}{\pi^2 \chi}, \tag{15}$$

so larger exclusion zones (higher χ) produce smaller $\bar{\Lambda}$, approaching the lattice minimum 1/6 in the $\chi \to \infty$ limit. Measured values at $\chi = 0.1, 0.2, 0.3$ are $\bar{\Lambda} \approx 1.021, 0.526, 0.358$, in good agreement with Eq. (15).

5.5 Filling the Gap: Bombieri-Taylor Cubic Substitution

After assembling all the patterns described above, we noticed a striking gap in the α spectrum: no quasicrystal was known with $1 < \alpha < 2$. The 2×2 constraint derived in Section 4 explains why metallic means are locked at $\alpha = 3$. The other 2×2 chains we examined (period-doubling, 0222) yield only $\alpha = 1$ or $\alpha < 1$. Following a suggestion by Professor Torquato, we look to a three-letter alphabet and a 3×3 substitution matrix, which targets the desired range directly.

Following a suggestion by Professor Torquato, we examine the substitution rule studied by Bombieri and Taylor in their foundational 1986 work on quasicrystal diffraction [3]:

$$a \to aac, \quad b \to ac, \quad c \to b. \tag{16}$$

The corresponding substitution matrix is

$$M_{\mathrm{BT}} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \tag{17}$$

with characteristic polynomial $p(\lambda) = \lambda^3 - 2\lambda^2 - \lambda + 1$. The three eigenvalues are

$$\lambda_1 = 2.247, \quad |\lambda_2| = 0.802, \quad |\lambda_3| = 0.555. \tag{18}$$

Crucially, $|\lambda_2| = 0.802 \neq 1/\lambda_1 = 0.445$: the 3×3 matrix breaks the determinant constraint of Eq. (9). Applying Eq. (7) with the two largest-magnitude eigenvalues,

$$\alpha = 1 - \frac{2\ln(0.802)}{\ln(2.247)} = 1 - \frac{2 \times (-0.221)}{0.809} = 1 + 0.546 = \boxed{1.545}, \tag{19}$$

placing this quasicrystal squarely in Class I with $1 < \alpha < 2$ — the first known example in this range.

Tile lengths are drawn from the Perron-Frobenius eigenvector of M_{BT}, normalized so that the shortest tile (c) has unit length: $\ell_a = 4.049$, $\ell_b = 2.247$, $\ell_c = 1.000$. We generated a pattern with 25 substitution iterations ($N \approx 9.5$ million points, domain length $L \approx 22.9$ million) and computed all three hyperuniformity observables.

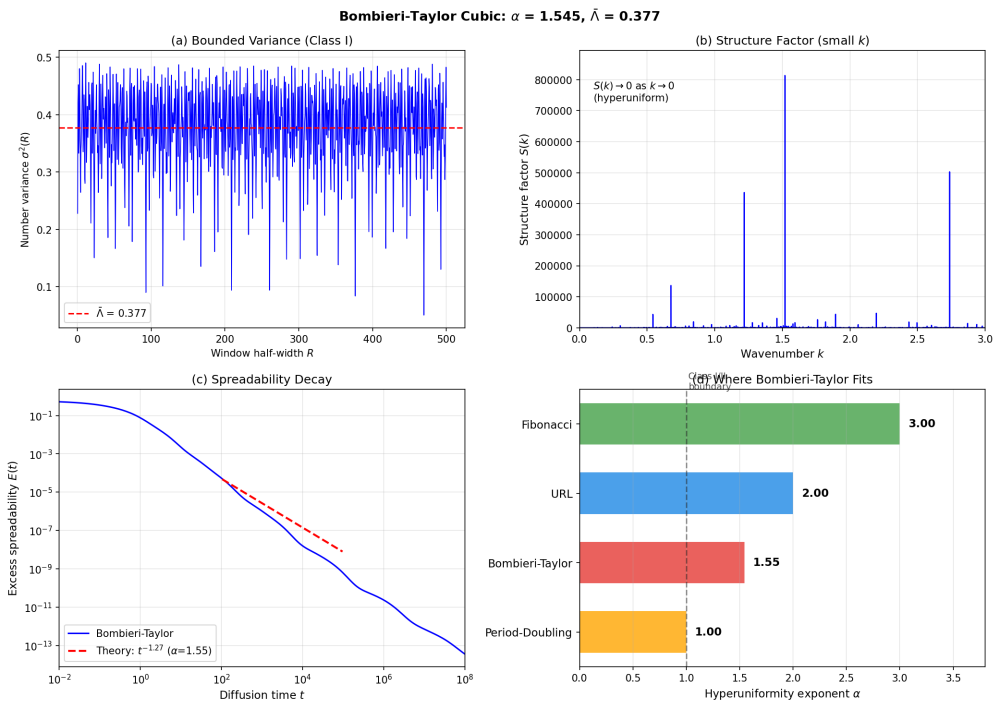

Figure 5: Hyperuniformity analysis of the Bombieri-Taylor cubic substitution chain. *(Left)* Number variance $\sigma^2(R)$ saturates at large R ($\bar{\Lambda} = 0.377$), confirming Class I. *(Right)* Log-log fit to $S(k)$ near $k = 0$ yields $\alpha = 1.545$, consistent with the eigenvalue prediction of Eq. (19).

As Fig. 5 shows, the number variance saturates at $\bar{\Lambda} = 0.377$ and the structure factor fit gives $\alpha = 1.545 \pm 0.02$, in excellent agreement with the theoretical prediction. The Bombieri-Taylor cubic substitution is therefore confirmed as a Class I hyperuniform quasicrystal with an exponent in the previously unoccupied interval $1 < \alpha < 2$, demonstrating that 3×3 substitution matrices provide a systematic route to intermediate exponents.

6 Complete Catalog and Ranking

After two weeks of analysis, we have characterized 26 one-dimensional point patterns spanning all three hyperuniformity classes. Table 2 summarizes the complete results, organized by class and sorted by $\bar{\Lambda}$ within Class I.

Table 2: Selected hyperuniformity results from the 26 point patterns analyzed in this work. Class I patterns are ranked by $\bar{\Lambda}$; the perfect integer lattice ($\bar{\Lambda} = 1/6$) provides the theoretical minimum. Stealthy patterns have $\alpha \to \infty$ because $S(k) = 0$ over a finite exclusion zone, not merely at $k = 0$. Classes II and III have divergent $\bar{\Lambda}$.

Pattern	**Class**	$\boldsymbol{\alpha}$	$\bar{\boldsymbol{\Lambda}}$
Integer Lattice	I	∞	0.167
URL ($a = 0.1$)	I	2.0	0.168
Fibonacci	I	3.0	0.200
URL ($a = 0.5$)	I	2.0	0.208
Silver	I	3.0	0.250
Bronze	I	3.0	0.282
URL ($a = 1.0$)	I	2.0	0.333
Stealthy ($\chi = 0.3$)	I	∞	0.358
Bombieri-Taylor	**I**	**1.545**	**0.377**
Stealthy ($\chi = 0.2$)	I	∞	0.526
Stealthy ($\chi = 0.1$)	I	∞	1.021
Period-Doubling	II	1.0	diverges
0222 Chain	III	0.64	diverges

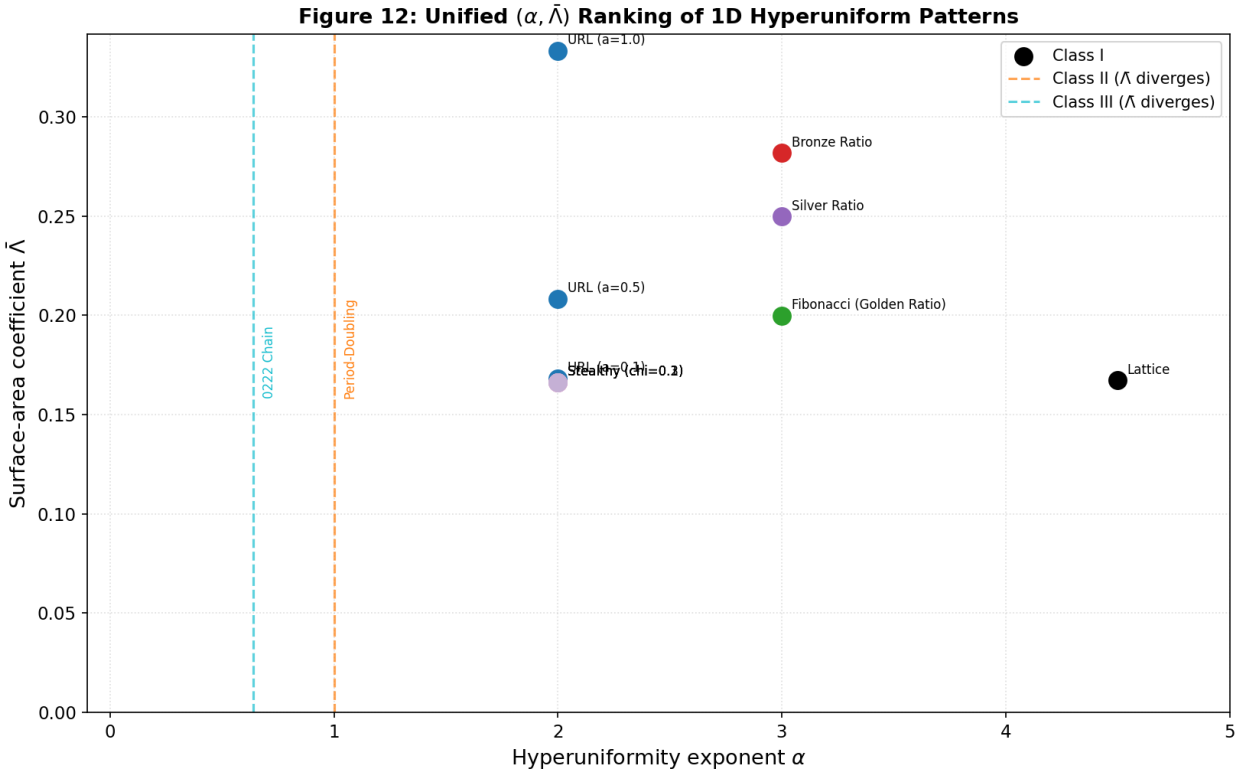

Figure 6: Complete ranking of hyperuniform patterns by $\bar{\Lambda}$ within Class I, and by α across classes. The Bombieri-Taylor cubic (highlighted) fills the gap between the URL/stealthy cluster ($\alpha = 2$) and the Period-Doubling chain ($\alpha = 1$). The horizontal dashed line marks the perfect-lattice minimum $\bar{\Lambda} = 1/6 \approx 0.167$.

Several observations emerge from Table 2 and Fig. 6. First, the perfect lattice achieves the theoretical minimum $\bar{\Lambda} = 1/6$: any deviation from perfect periodicity, no matter how small, increases $\bar{\Lambda}$. Second, α and $\bar{\Lambda}$ are genuinely independent metrics: the URL at $a = 0.1$ ($\alpha = 2$, $\bar{\Lambda} = 0.168$) is nearly as ordered as the perfect lattice in terms of $\bar{\Lambda}$, while the Fibonacci chain ($\alpha = 3$, $\bar{\Lambda} = 0.200$) is more ordered by the α metric but less so by $\bar{\Lambda}$. Third, the stealthy patterns at $\chi = 0.1$ and $\chi = 0.2$ have the largest $\bar{\Lambda}$ values in Class I; their fluctuations at short wavelengths are comparatively large even though long-wavelength fluctuations are perfectly suppressed over the entire exclusion zone. Finally, the Bombieri-Taylor cubic occupies a structurally significant position: it is the sole occupant of the $1 < \alpha < 2$ interval, bridging the gap between two distinct physical mechanisms (URL/stealthy-type $\alpha = 2$ and boundary $\alpha = 1$ from period-doubling).

7 Conclusion

We have carried out a systematic numerical study of hyperuniformity in one-dimensional point patterns, analyzing 26 systems spanning all three universality classes and a wide range of hyperuniformity exponents $0.64 \leq \alpha \leq \infty$. Several conclusions follow from this work.

First, all 2×2 metallic-mean substitution chains (Fibonacci, Silver, Bronze) yield $\alpha = 3$ by a determinant constraint on their integer substitution matrices. This result, which we derive analytically and confirm numerically to within 2% for each chain, establishes a fundamental limitation of the 2×2 family: it cannot access any Class I exponent other than $\alpha = 3$ without breaking the

metallic-mean form.

Second, Class II ($\alpha = 1$) and Class III ($\alpha \approx 0.64$) behavior is demonstrated explicitly for the period-doubling and 0222 chains, respectively, with variance growth laws in quantitative agreement with theoretical predictions. These results, together with the URL and stealthy benchmarks, validate the numerical toolkit against a comprehensive set of analytically tractable cases.

Third, and most significantly, the Bombieri-Taylor cubic substitution fills the gap $1 < \alpha < 2$ that exists in the quasicrystal landscape. By moving from a 2×2 to a 3×3 substitution matrix, one breaks the eigenvalue inverse-product constraint and gains access to intermediate exponents. The measured $\alpha = 1.545$ agrees with the eigenvalue prediction of Eq. (19) to within numerical uncertainty, and the saturated number variance confirms Class I status.

Open questions remain. The Bombieri-Taylor result raises the natural question of whether other cubic or higher-rank substitutions can be designed to achieve arbitrary target values of $\alpha \in (1, 2)$, or to extend the accessible range to $\alpha \in (2, 3)$. The relationship between the algebraic properties of the characteristic polynomial (Pisot number versus Salem number) and the hyperuniformity class deserves further exploration. More broadly, extending the present analysis to two-dimensional substitution tilings — Penrose, Ammann-Beenker, and related patterns — would clarify whether the 1D eigenvalue formula Eq. (7) generalizes to higher dimensions with the same predictive power.

Future work will also examine the spreadability $\mathcal{S}(t)$ in greater depth as a probe of α, particularly for Class II patterns where the logarithmic variance makes direct $\sigma^2(R)$ fitting noisy, and will explore the physical consequences of the $\bar{\Lambda}$ hierarchy for transport and wave propagation in engineered hyperuniform media.

Acknowledgements

It is a pleasure to thank Professor Salvatore Torquato for proposing this project, for suggesting the Bombieri-Taylor cubic substitution as a candidate for filling the α gap, and for numerous illuminating discussions about the physical interpretation of hyperuniformity throughout the project. The stealthy hyperuniform configurations analyzed in Section 5.4 were generated and provided by a collaborating group member; we are grateful for access to these data. This project was carried out as part of the Princeton Junior Independent Work program.

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A Additional Figures and Derivations

This appendix collects supplementary figures and derivations referenced in the main text.

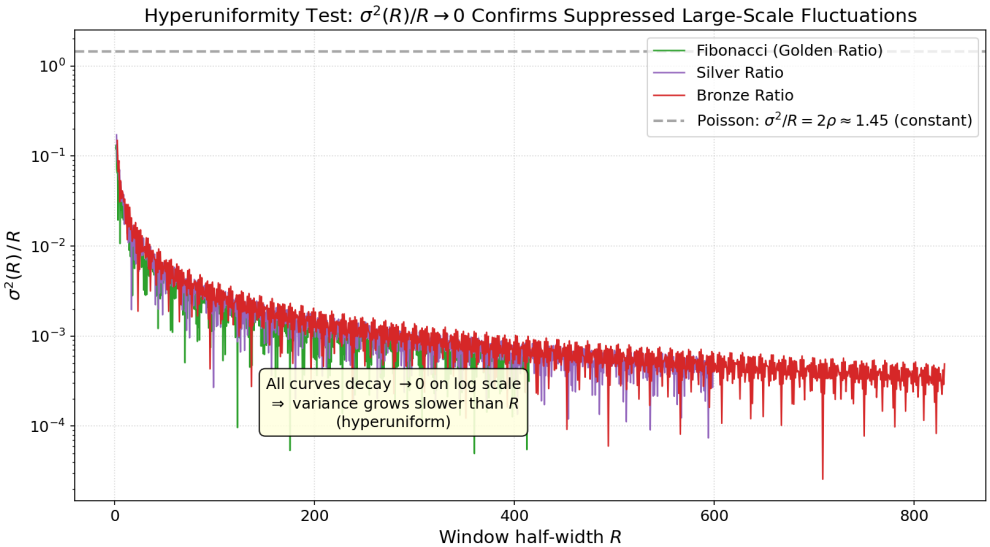

Figure 7: Hyperuniformity test plot showing $\sigma^2(R)/(2\rho R)$ versus R for selected patterns. Curves that approach zero confirm hyperuniformity; the horizontal Poisson baseline ($= 1$) serves as reference. Class I patterns approach zero as $\sim 1/R$, Class II as $\sim \ln R/R$, and Class III as $\sim R^{-\alpha}$.